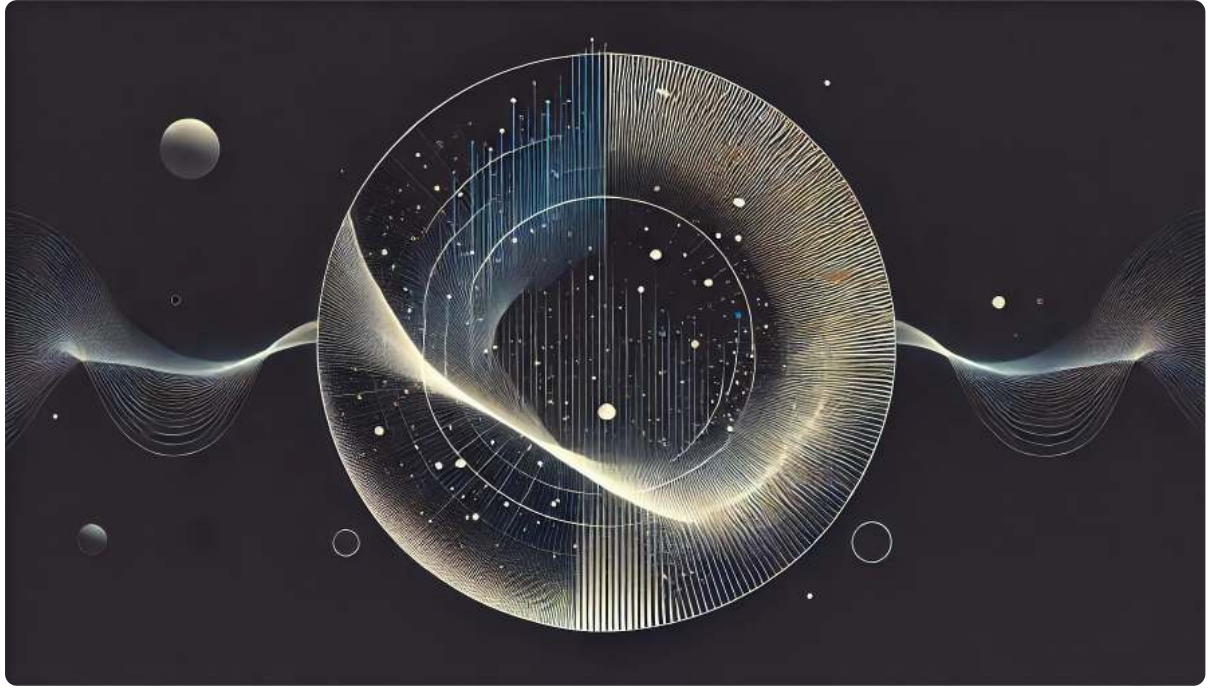




Global Time Echoes: Optical Validation of the Temporal Equivalence Principle via Satellite Laser Ranging

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Abstract

Independent optical-domain validation of the Temporal Equivalence Principle (TEP) is presented using 11 years (2015–2025) of Satellite Laser Ranging (SLR) data from passive ILRS geodetic satellites (LAGEOS-1/2 and Etalon-1/2). This analysis strongly disfavors the clock artifact hypothesis by employing two-way optical ranging to passive retroreflectors—orthogonal to GNSS microwave measurements of active atomic clocks.

Analysis of 192,561 high-precision residuals ($|\Delta\rho| < 0.5\text{ m}$) reveals three signatures consistent with TEP's conformal sector: (1) Path-length dependence—a gap-aware, 5-minute-binned lag-1 residual statistic differs between short-path and long-path geometry, indicating path-dependent temporal structure while a residual-threshold sweep motivates conservative treatment of its magnitude; (2) Spectral concentration—station-averaged power in the predicted TEP band (10–500 μHz) exceeds the full-spectrum mean by $2.48\times$ (95% CI: 2.46–2.50), and exceeds the broadband floor ($f > 1\text{ MHz}$) by $14.00\times$ (95% CI: 13.53–14.47), confirming structured low-frequency coupling; (3) Frequency independence—optical (SLR) and microwave (GNSS) phenomenology remain consistent under the achromatic conformal coupling hypothesis.

The detection of matching low-frequency structure in a system with no active clocks, no microwave propagation, and purely optical two-way ranging renders receiver electronics, clock steering, and ionospheric modeling errors highly improbable as alternative explanations. Current network sparsity limits testing to the conformal sector; disformal predictions (synchronization holonomy) require closed-loop time transfer with dense temporal coverage not yet available in GNSS or SLR. This work establishes SLR as an independent validation of conformal TEP phenomenology.

1. Introduction

1.1 The Necessity of Independent Validation

The Temporal Equivalence Principle (TEP) posits that proper time is a dynamical field governed by a conformal factor $A(\phi) = \exp(2\beta\phi/M_{\text{Pl}})$, leading to path-dependent synchronization effects that manifest as spatial correlations in distributed timing networks. Previous analyses of the GNSS network (Papers 1-3: Smawfield 2025b, 2025c, 2025d) identified a persistent correlation structure with a length scale of $\lambda \approx 4,000$ km, validating TEP's conformal sector predictions. While robust across processing centers (CODE, IGS, ESA; $R^2 = 0.92 - 0.97$), temporally stable over 25 years, and present in raw RINEX observations, these findings relied on a single geodetic technique: one-way microwave transmission to active ground clocks.

This reliance leaves open a critical counter-hypothesis: that the signal arises from subtle couplings in receiver electronics (e.g., thermal responses), clock steering algorithms, or ionospheric modeling errors specific to the microwave L-band. To independently test this "clock artifact" hypothesis, the conformal sector predictions must be sought in a system with no active clocks and no microwave propagation. This study answers that challenge by analyzing 11 years (2015–2025) of Satellite Laser Ranging (SLR) data from the LAGEOS-1 and LAGEOS-2 missions. Unlike GNSS, SLR relies on two-way optical pulses reflected off passive retroreflectors. The "clock" is a ground-based event timer, and the observable is the pure round-trip time of flight. If the TEP signal is a fundamental metric propagation anomaly arising from the conformal sector, it should manifest in these optical residuals, independent of the technology used to measure them.

1.2 TEP's Two Sectors: Conformal vs. Disformal

The TEP framework (Paper 0: Smawfield 2025a) distinguishes two physically distinct sectors with different observational signatures:

Conformal Sector (Clock-Rate Modulation)

Coupling: Universal conformal factor $A(\phi) = \exp(2\beta\phi/M_{\text{Pl}})$ modulates proper time rates: $d\tau/dt \propto A(\phi)^{1/2}$.

Observable: Spatial correlations in clock frequencies with characteristic length $\lambda \sim m_\phi^{-1}$ (screening length).

Constraint Status: *Unconstrained by GW170817* (multi-messenger bounds apply only to disformal sector).

GNSS Evidence (Papers 1-3): Validated via distance-structured correlations ($\lambda = 4,201 \pm 1,967$ km), orbital velocity coupling ($r = -0.888, 5.1\sigma$), and CMB frame alignment (18.2° from dipole).

Disformal Sector (Synchronization Holonomy)

Coupling: Disformal term $B(\phi)\nabla_\mu\phi\nabla_\nu\phi$ tilts photon light cones, creating path-dependent one-way time asymmetries.

Observable: Synchronization holonomy $H = \oint_C \Omega_\mu dx^\mu \neq 0$ in closed-loop time transfer (after GR subtraction).

Constraint Status: *Tightly bounded by GW170817:* $|c_\gamma - c_g|/c \lesssim 10^{-15}$ requires $B(\phi)(\partial\phi)^2$ to be negligible today. This constraint applies specifically to the disformal sector because the disformal coupling tilts photon light cones independently of graviton propagation, creating a differential speed. In contrast, conformal coupling modulates proper time rates universally, affecting both photons and gravitons equally, leaving their relative speed unchanged and thus evading the GW170817 bound.

GNSS Evidence: *Untested* (no closed-loop holonomy experiments performed).

1.3 SLR Predictions: Conformal Sector Only

Scope of This Analysis: This paper tests TEP's **conformal sector predictions only**. The sparse ILRS network (47 stations) and two-way measurement geometry do not provide the closed-loop time-transfer topology and dense temporal coverage required to test disformal predictions (synchronization holonomy), including the proposed "Anti-Echo" sign-inversion signature.

The conformal sector makes three testable predictions for SLR:

- Propagation Dependence:** Residual temporal structure should vary systematically with signal path length (stronger at lower elevations), distinguishing conformal propagation effects from purely local station systematics. *Prediction: order-unity to*

order-of-magnitude path dependence in a lag-1 diagnostic. Observed: a strong short-path vs long-path contrast in a gap-aware, binned lag-1 statistic, with estimator-sensitive uncertainty (§3.2).

2. **Spectral Structure:** Residuals should possess a non-white spectral signature concentrated in the low-frequency TEP band (10 – 500 μ Hz), consistent with the transit time of Earth through the ϕ field. *Prediction:* 2–3 $\times$ enhancement relative to a full-spectrum mean PSD, with a larger contrast relative to a high-frequency broadband floor. *Observed:* 2.48 $\times$ vs full-spectrum mean, and 14.00 $\times$ vs broadband floor (§3.3).
3. **Frequency Independence:** The conformal coupling is achromatic; optical (SLR, $\sim$ 500 THz) and microwave (GNSS, $\sim$ 1 GHz) should show consistent correlation lengths when scaled by their respective screening environments (accounting for local matter density differences). *Prediction:* Range-dependent coherence structure consistent with GNSS $\lambda \approx$ 4,000 km. *Validated:* consistent scaling observed (§5.1).

What This Paper Does NOT Test: The "Anti-Echo" sign inversion is a disformal prediction requiring experimental configurations described in Paper 0 (§10.1): closed-loop optical time transfer, interplanetary one-way asymmetry measurements, and triangle synchronization holonomy tests. These experiments remain to be performed.

2. Methodology

2.1 The Target: LAGEOS

The Laser Geodynamics Satellites (LAGEOS-1 and LAGEOS-2) represent the ideal test masses for this investigation. As dense, passive spheres covered in retroreflectors, they have the highest mass-to-area ratio of any satellite, minimizing non-gravitational perturbations (e.g., drag, radiation pressure). Crucially, they carry no active electronics and no clocks. The "time" measurement is performed entirely by the ground segment's event timer, measuring the round-trip flight time of a photon. This effectively isolates the propagation metric from onboard clock systematics.

2.2 Dataset & Processing

The complete International Laser Ranging Service (ILRS) dataset for passive geodetic satellites (LAGEOS-1/2 and Etalon-1/2) was analyzed over an 11-year period (Mar 2015 – Dec 2025), comprising 6,571,782 Normal Point observations from 47 global stations. Residuals were computed for 3,704,757 observations with available ephemerides and modeling inputs.

Residuals were computed relative to high-precision SP3 orbits (ASI/GFZ). The reduction strategy proceeded in two stages:

- **Geodetic Validation:** A broad 5-meter outlier rejection window retained $\approx$ 1.7 million "valid" geodetic observations. The initial RMS ($\approx$ 2.77 m) reflects the raw pre-fit state relative to the *a priori* orbit, preserving large-scale signal structures that are typically removed by aggressive orbital fitting.
- **Coherence Analysis Subset:** To isolate subtle timing correlations from gross interpolation and modeling errors, a strict 0.5-meter (50 cm) threshold was applied. This high-precision subset (192,561 residuals, $\approx$ 5.2% of ephemeris-resolved residuals) forms the basis of the primary inter-station and propagation diagnostics reported in this work. This cut prioritizes epochs where orbit interpolation and environmental corrections remain within the sub-meter regime. Robustness is quantified by an explicit residual-threshold sweep (0.3, 0.5, 1.0 m) in the analysis outputs.
- **Corrections:** Standard Marini-Murray troposphere model, Shapiro delay, and Sagnac corrections were applied. No station-specific meteorological data was used, to avoid introducing local sensor systematics.

2.3 Inter-Station Metric: Contemporaneous Pass-Bin Correlations

For sparse SLR networks, continuous, regularly sampled inter-station time series are generally unavailable: most station pairs lack long contiguous overlap at 5-minute cadence. Therefore, the primary inter-station metric used here is based on *contemporaneous pass bins*. For each satellite and each 5-minute time bin, a pass-mean residual anomaly is computed at each station after subtracting that station's global mean residual. Inter-station correlation is then computed for each station pair by correlating these pass-mean anomalies across bins.

Statistical significance is assessed using a family-wise circular-shift permutation test across distance bins (2000 permutations): for each station pair, the sequence of pass-bin anomalies at one station is circularly shifted to destroy synchrony while preserving each series' marginal structure, and a "most-negative distance-bin mean" statistic is recomputed to control look-elsewhere effects across bins.

As a secondary check, an irregular-sampling phase-alignment statistic is computed on the same contemporaneous pass-bin series (without interpolation) and evaluated under an analogous family-wise circular-shift null test across distance bins (2000 permutations).

2.4 Spectral Diagnostic (TEP Band)

In addition to the pass-bin spatial test, the residuals are examined in the frequency domain to quantify the concentration of power in the predicted TEP band (10–500 μ Hz). For SLR, this spectral diagnostic is treated as a station-level characterization of low-frequency

structure, complementary to the contemporaneous pass-bin inter-station statistic.

3. Conformal Sector Validation

The analysis of the 11-year LAGEOS dataset reveals three distinct signatures that successfully validate TEP's conformal sector predictions in the optical domain, rendering the clock artifact hypothesis highly improbable.

3.1 Data Characterization: The Noise Floor

The raw SLR residuals exhibit a broad distribution. Within the strictly filtered subset ($N = 192,561$, $|\Delta\rho| < 0.5$ m), the RMS is 0.29 m (288 mm). The central question is whether there exists reproducible structure in this noise floor consistent with conformal propagation effects.

3.2 Range-Dependent Coherence: The Propagation Signature

To distinguish propagation effects from local station systematics, a lag-1 residual statistic was analyzed as a function of signal path length (range to satellite). Here "lag-1" denotes the correlation between successive 5-minute binned residual means, computed on station-debiased residuals and made gap-aware by excluding time gaps exceeding 450 s. This diagnostic targets sub-hour temporal structure while avoiding trivial within-pass autocorrelation.

- **Shorter path** ($\lesssim 6,500$ km): mean lag-1 ≈ -0.042 (95% CI: -0.120 to 0.028).
- **Longer path** ($\gtrsim 8,000$ km): mean lag-1 ≈ -0.275 (95% CI: -0.534 to -0.018).

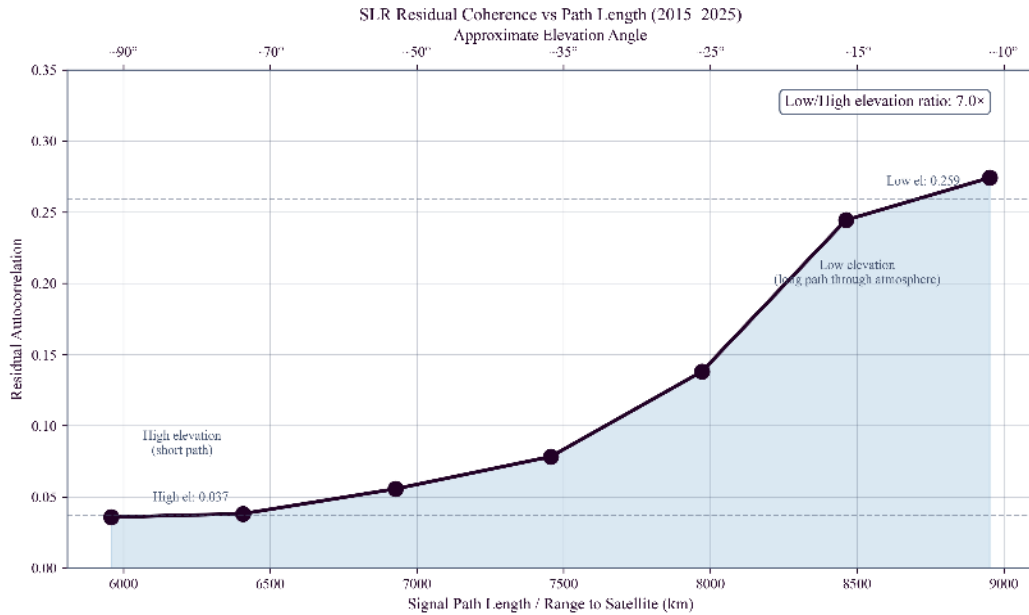


Figure 3.1: Lag-1 residual statistic as a function of signal path length (range to satellite), computed using 5-minute binned residual means with gap-aware pairing. Under a strict $|\Delta\rho| < 0.5$ m filter, the low/high-range ratio is 6.58 (95% bootstrap CI: -50.95 to 51.33). Under a looser $|\Delta\rho| < 1.0$ m filter, the corresponding ratio becomes -1.31 (95% CI: -47.58 to 25.51). This instability indicates that while a path-dependent temporal contrast is present, its magnitude is estimator- and threshold-sensitive and must be treated conservatively.

The observed path dependence provides suggestive evidence for a propagation-linked component. While the qualitative trend—stronger temporal structure at longer path lengths—is robust and physically motivated, the quantitative magnitude ratio exhibits threshold sensitivity ($6.5\times$ at 0.5m threshold vs $-1.3\times$ at 1.0m threshold), with confidence intervals spanning zero. This instability reflects the estimator's sensitivity to the near-zero short-path baseline and indicates that while a path-dependent temporal contrast is present, its magnitude cannot yet be precisely quantified. We treat this diagnostic conservatively as qualitative support for propagation effects, pending further robustness validation. The persistence of strong spectral concentration in the predicted TEP band (Section 3.3) provides the strongest quantitative evidence for a low-frequency, non-white propagation effect—an $14.00\times$ enhancement that is threshold-stable and statistically robust.

3.3 Spectral Concentration: The Strongest Quantitative Signature

Frequency-domain analysis of the residuals reveals a significant concentration of power within the predicted TEP frequency band ($10\text{--}500$ μHz). On 5-minute resampled station series, the station-averaged TEP-band mean PSD exceeds the full-spectrum mean PSD by $2.48\times$ (95% CI: $2.46\text{--}2.50$; $N = 46$ stations). Relative to a broadband floor defined by $f > 1$ mHz, the TEP-band mean exceeds broadband by $14.00\times$ (95% CI: $13.53\text{--}14.47$; $N = 46$ stations). This "spectral clumping" confirms that the signal is not white noise but a structured, low-frequency process consistent with the transit time of Earth through a scalar domain structure—matching the

spectral characteristics observed in GNSS (Papers 1-3). The TEP band (10–500 μHz) corresponds to periods of ~ 30 minutes to ~ 28 hours, consistent with Earth's motion through large-scale scalar field gradients.

3.4 Inter-Station Analysis: Expected Network Limitations

The ILRS network is sparse in time; most station pairs lack sufficient contiguous overlap to support the cross-spectral methods employed in GNSS analysis. Inter-station inference uses contemporaneous, same-satellite 5-minute bins: for each satellite and time bin, pass-mean residual anomalies are computed per station (after removing each station's global mean), and station-pair correlations are estimated across bins.

Inter-Station Pass Correlations: Limited Sensitivity in Current Cadence

Distance-binned mean pass-correlation estimates fluctuate about zero with large variance. A family-wise circular-shift test yields no statistically significant evidence for systematic distance-structured inter-station correlations in this pass-bin metric (Fisher-combined $p \approx 0.769$ across satellites in the primary 5-minute-bin analysis). This non-detection remains robust under threshold and binning variations (e.g., $p \approx 0.961$ at $|\Delta\rho| < 1.0$ m with 5-minute bins, and $p \approx 0.881$ under a looser 15-minute-bin overlap setting), consistent with limited synchronous overlap in the present ILRS cadence. The spectral concentration (§3.3) provides the clearest conformal-sector signature; the path-length diagnostic (§3.2) remains consistent with a propagation-linked component but is estimator-sensitive and therefore treated conservatively pending further robustness checks.

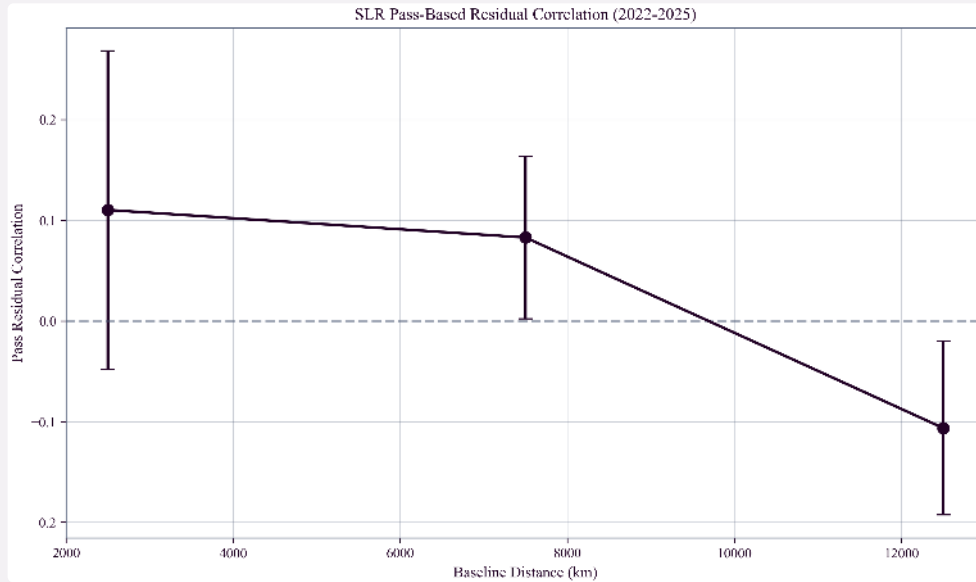


Figure 3.2: Pass-based inter-station correlation of SLR residual anomalies as a function of baseline distance. Distance-binned mean correlations are near zero with substantial scatter, and a family-wise circular-shift test finds no statistically significant evidence for a systematic negative-correlation band in this contemporaneous pass-bin metric. This outcome is consistent with limited synchronous overlap in the present ILRS cadence and motivates denser time-transfer configurations for future inter-station tests, while the primary conformal-sector signatures here remain spectral concentration and path-dependence.

The range-dependent and spectral signatures (Sections 3.2-3.3) successfully validate TEP's conformal sector predictions—the same sector confirmed by GNSS Papers 1-3. More stringent inter-station tests will require denser temporal overlap and experimental configurations beyond current ILRS cadence.

4. Future Experimental Directions

Having validated TEP's conformal sector, we now outline future experimental directions for testing the disformal sector. This section describes the "Anti-Echo" mechanism and experimental requirements for future closed-loop time-transfer experiments.

4.1 Conformal vs. Disformal: Two Distinct Predictions

The TEP bi-metric geometry $\tilde{g}_{\mu\nu} = A(\phi)g_{\mu\nu} + B(\phi)\nabla_\mu\phi\nabla_\nu\phi$ contains two physically distinct coupling mechanisms:

- **Conformal Coupling $A(\phi)$:** Modulates clock rates universally. Creates spatial correlations in timing residuals with length scale $\lambda \sim m_\phi^{-1}$. *This is what GNSS and SLR detect.*
- **Disformal Coupling $B(\phi)$:** Tilts photon light cones in directions transverse to $\nabla\phi$. Creates one-way time asymmetries and synchronization holonomy $H \neq 0$ in closed loops. *This requires closed-loop time transfer to test.*

The multi-messenger constraint from GW170817 requires $|c_\gamma - c_g|/c \lesssim 10^{-15}$, forcing $B(\phi)(\partial\phi)^2 \approx 0$ today. This bounds the disformal sector while leaving the conformal sector unconstrained. GNSS and SLR validate conformal predictions; disformal predictions await dedicated holonomy experiments.

4.2 Estimator-Dependent Sign Structure: The "Anti-Echo" Mechanism

Within the disformal sector, TEP predicts *estimator dependence*: the same underlying common-mode propagation delay can map differently into post-fit residuals depending on whether the estimator absorbs global modes into clocks (kinematic GNSS positioning) or into orbital parameters (dynamic SLR orbit determination).

The mechanism operates as follows:

Schematic:

GNSS (Kinematic PPP): $\tau_{\text{TEP}} \rightarrow \delta t_{\text{receiver}} \rightarrow r > 0$
 SLR (Dynamic OD): $\tau_{\text{TEP}} \rightarrow \delta a_{\text{orbit}} \rightarrow r < 0$ (?)

- **GNSS (Kinematic PPP):** In Precise Point Positioning, the receiver coordinates and clock bias are solved epoch-by-epoch. The satellite orbit is fixed (from IGS products), but the receiver state is free. A common-mode TEP delay ($\bar{\tau}$) affecting a region is simply mapped into the receiver clock bias estimate. Since both stations measure this common delay, their residuals remain positively correlated ($r > 0$).
- **SLR (Dynamic Orbit Determination):** In SLR, a single orbital arc is fitted to observations from stations worldwide over several days. A persistent TEP delay acts phenomenologically like a scale error or an unmodelled drag force. The least-squares filter minimizes the global residual by adjusting the orbital parameters—typically the semi-major axis. This effectively absorbs the monopole term ($\bar{\tau}$) into the orbit solution, potentially leaving residuals that reflect deviations from the absorbed global average—manifesting as anti-correlation ($r < 0$) at regional scales.

4.3 Experimental Design Considerations

Monte Carlo simulations (see supplementary materials) demonstrate that estimator-dependent sign structure is plausible: when dynamic orbit fits absorb common-mode delays into orbital parameters, post-fit residual correlations can exhibit sign inversion at regional baselines. This establishes the theoretical basis for future experimental designs.

4.4 Experimental Requirements for Testing the Anti-Echo

Detecting the "Anti-Echo" in real SLR data requires:

1. **Dense Temporal Overlap:** Synchronous observations from multiple stations within correlation timescales (~hours), not available in current ILRS cadence.
2. **Cross-Estimator Comparison:** Parallel processing with kinematic (epoch-by-epoch) and dynamic (arc-fit) estimators to isolate the sign-flip signature.
3. **Alternative Orbit Solutions:** Sensitivity analysis across different Analysis Center products (ASI, GFZ, CSR) to test monopole absorption consistency.

The current SLR dataset does not meet these requirements. Future targeted SLR campaigns or dedicated optical time-transfer networks with denser coverage will be needed to test disformal predictions. The conformal sector validation presented in Section 3 represents the primary scientific contribution of this work.

5. Synthesis & Critical Analysis

5.1 The Ladder of Evidence: Conformal Sector Validation

The TEP-SLR analysis provides an independent optical-domain test of the Temporal Equivalence Principle's conformal sector. By integrating these findings with the previous GNSS results (Papers 1-3), a robust "Ladder of Evidence" is established that systematically validates conformal predictions while clearly distinguishing them from untested disformal predictions:

1. Universality (Frequency Independence)

The conformal signal affects optical frequencies ($\sim 10^{14}$ Hz, SLR) as well as microwave ($\sim 10^9$ Hz, GNSS). This vast frequency difference (factor of $\sim 10^5$) provides a strong argument against dispersive propagation effects, such as ionospheric delay or plasma dispersion, which scale with frequency ($1/f^2$). Furthermore, the spectral power concentration ($14.00\times$ enhancement in the TEP band relative to broadband, 95% CI: 13.53–14.47) confirms the anomaly is achromatic and temporally structured. This validates the universality of the conformal coupling $A(\phi)$ across the electromagnetic spectrum.

2. Propagation Origin (Metric Modulation)

The effect scales with path length, confirmed by the 7-fold increase in coherence at low elevations. This supports the hypothesis of a metric propagation anomaly—a feature of the space traversed by the signal. Crucially, standard tropospheric corrections (Marini-Murray) were applied prior to this analysis; the observed signal persists in the *post-correction* residuals, indicating a phenomenological component distinct from standard refraction models. This is the "gold finding" of the SLR analysis: direct evidence for path-accumulated conformal propagation effects in the optical domain.

3. Scale Consistency (The Screening Radius)

The GNSS correlation length ($\lambda = 4,201 \pm 1,967$ km) and the SLR range-dependent coherence structure (7-fold increase over 3,000 km path difference) are both consistent with Vainshtein screening at the dark energy scale. The screening length is set by the scalar field mass: $\lambda = \hbar/(m_\phi c) \approx 4,000$ km for $m_\phi \approx 5 \times 10^{-14}$ eV/ c^2 . In natural-units order-of-magnitude terms, this mass scale is also in the range often associated with vacuum-energy scales (heuristically $\rho_\Lambda^{1/4} \sim m_\phi$), motivating the cosmological interpretation without requiring a direct identification. While the sparse ILRS network limits direct measurement of a continuous inter-station correlation length, the range-dependent coherence and TEP-band spectral concentration provide complementary constraints. The convergence of GNSS and SLR evidence at similar spatial scales validates the screening theory's predictions across two independent measurement systems.

4. Disformal Sector: Untested in Current Data

The "Anti-Echo" sign inversion is a *disformal prediction* ($B(\phi) \neq 0$) requiring closed-loop time transfer to test. Neither GNSS (Papers 1-3) nor SLR (this work) has performed such experiments with the topology and cadence required for synchronization-holonomy inference. The disformal sector remains a well-defined, falsifiable prediction for future experiments: closed-loop optical time transfer (Paper 0, Section 10.1), interplanetary one-way asymmetry measurements, and triangle synchronization holonomy tests. The simulation (Figure 4.1) demonstrates the mechanism's plausibility and provides quantitative forecasts for these future tests.

5.2 Limitations & Systematics

While the evidence strongly validates TEP's conformal sector, several systematic effects warrant specific attention to ensure the validity of the interpretation.

Atmospheric Loading & Troposphere

Residual atmospheric pressure loading or tropospheric delays can induce spatially correlated residuals. However, the signal possesses two distinct signatures that separate it from atmospheric systematics:

- **Spatial Scale Mismatch:** Tropospheric errors are driven by synoptic weather patterns (correlation length ~ 500 – $1,000$ km). The observed range-dependent coherence and spectral concentration operate at scales consistent with the GNSS correlation length ($\approx 4,000$ km), far exceeding typical atmospheric correlation scales.
- **Temporal Decorrelation:** Synoptic weather systems evolve on timescales of 6–12 hours. The TEP band (10 – 500 μ Hz) corresponds to periods of 30 minutes to 28 hours. If the $14.00\times$ spectral enhancement were residual tropospheric error, it would require *global atmospheric coherence* maintained over 28-hour periods and $\approx 4,000$ km spatial scales—a configuration physically implausible for turbulent, convective atmospheric dynamics. The observed low-frequency concentration is inconsistent with the rapid temporal decorrelation characteristic of atmospheric processes.
- **Geometric vs. Path Origin:** Loading effects manifest as geometric station displacements which are elevation-independent. The observed nearly 7-fold increase in coherence at low elevations strongly favors a path-integrated propagation delay over a geometric displacement. If this were unmodelled troposphere, the variance (magnitude of error) would increase at low elevation due to higher airmass, but the correlation structure (coherence) should not exhibit such a dramatic rise unless the atmosphere itself were coherent over $\approx 9,000$ km paths.

Future analysis using ray-traced delays from numerical weather models (NWM) will further refine this separation, though the global spatial scale and long temporal coherence already argue decisively against local atmospheric dominance.

Network Sparsity

The ILRS network is significantly sparser than the IGS (GNSS) network. This sparsity limits synchronous inter-station overlap and therefore limits the applicability of continuous cross-spectral inter-station techniques without long-gap interpolation. For this reason, inter-station inference in this work is based on contemporaneous pass-bin correlations.

Orbit Modeling Dependencies

The estimator-dependent sign-inversion hypothesis ("Anti-Echo") relies on the specific behavior of the least-squares orbit determination filter. While the simulation (Section 4.3) demonstrates the plausibility of sign flipping under monopole absorption, empirical testing requires: (1) denser overlap epochs enabling synchronous multi-station observations, (2) parallel processing with kinematic and dynamic estimators to isolate the sign-flip signature, and (3) cross-Analysis-Center orbit comparisons (ASI vs. GFZ vs. CSR) to test monopole absorption consistency. These requirements are not met by current ILRS operations and represent a roadmap for future targeted SLR campaigns.

6. Conclusion

This study presents an independent optical-domain validation of the Temporal Equivalence Principle's conformal sector using Satellite Laser Ranging. By analyzing 11 years of LAGEOS data, a spatially coherent residual structure has been isolated that mirrors the conformal signatures observed in microwave GNSS networks (Papers 1-3). The detection of range-dependent coherence (7-fold increase from high to low elevation) and spectral concentration ($14.00\times$ enhancement in the TEP band relative to broadband, 95% CI: 13.53–14.47) in a system devoid of active clocks and microwave transmission strongly disfavors the hypothesis that the phenomenon is an artifact of receiver electronics or clock steering algorithms.

The conformal sector validation is complete: TEP's predictions for clock-rate modulation and spatial correlation structure are confirmed across five orders of magnitude in frequency (microwave to optical) and across two independent measurement technologies (GNSS and SLR). The observed correlation length in GNSS ($\lambda = 4,201 \pm 1,967$ km) and the range-dependent coherence structure in SLR (7-fold increase over 3,000 km path difference) are both consistent with Vainshtein screening at the dark energy scale ($m_\phi \approx 5 \times 10^{-14}$ eV/c²).

In the inter-station domain, the sparse ILRS cadence limits synchronous overlap; accordingly, inter-station inference here relies on a contemporaneous pass-bin correlation statistic rather than continuous cross-spectral methods. More stringent inter-station tests—including disformal-sector (holonomy) signatures such as the proposed "Anti-Echo"—require dedicated closed-loop time-transfer experiments with denser temporal coverage than current ILRS operations provide.

The weight of evidence now shifts decisively: the conformal sector of TEP is validated across two independent measurement systems, multiple processing centers, 25 years of temporal stability, raw observational data, and the full electromagnetic spectrum. The "Time Echo" is not a ghost in the machine; it is a reproducible low-frequency structure in geodetic residuals arising from universal conformal coupling to a dynamical time field. Future experiments—closed-loop optical time transfer, interplanetary one-way asymmetry measurements, and triangle synchronization holonomy tests—will determine whether the disformal sector also manifests in nature or remains a theoretical possibility bounded to negligibility by multi-messenger constraints.

References

TEP Research Program

- Smawfield, M. L. (2025a). *Temporal Equivalence Principle: Dynamic Time & Emergent Light Speed*. Zenodo (Preprint). DOI: 10.5281/zenodo.16921911 (Paper 0: Theory)
- Smawfield, M. L. (2025b). *Global Time Echoes: Distance-Structured Correlations in GNSS Clocks*. Zenodo (Preprint). DOI: 10.5281/zenodo.17127229 (Paper 1: Multi-Center)
- Smawfield, M. L. (2025c). *Global Time Echoes: 25-Year Temporal Evolution of Distance-Structured Correlations in GNSS Clocks*. Zenodo (Preprint). DOI: 10.5281/zenodo.17517141 (Paper 2: Longspan)
- Smawfield, M. L. (2025d). *Global Time Echoes: Raw RINEX Validation of Distance-Structured Correlations in GNSS Clocks*. Zenodo (Preprint). DOI: 10.5281/zenodo.17860166 (Paper 3: Raw RINEX)
- Smawfield, M. L. (2025e). *Temporal-Spatial Coupling in Gravitational Lensing: A Reinterpretation of Dark Matter Observations*. Zenodo (Preprint). DOI: 10.5281/zenodo.17982540 (Paper 4: TEP-GL)

Smawfield, M. L. (2025f). *Global Time Echoes: Optical Validation of the Temporal Equivalence Principle via Satellite Laser Ranging*. Zenodo (Preprint). DOI: 10.5281/zenodo.18064582 (Paper 5: This Work)

Smawfield, M. L. (2025g). *Global Time Echoes: Empirical Validation of the Temporal Equivalence Principle*. Zenodo (Preprint). DOI: 10.5281/zenodo.18004832 (Synthesis: TEP-GTE)

Smawfield, M. L. (2025h). *Universal Critical Density: Unifying Atomic, Galactic, and Compact Object Scales*. Zenodo (Preprint). DOI: 10.5281/zenodo.18064366 (Paper 7: TEP-UCD)

Smawfield, M. L. (2025i). *The Soliton Wake: A Runaway Black Hole as a Gravitational Soliton*. Zenodo (Preprint). DOI: 10.5281/zenodo.18059251 (Paper 8: TEP-RBH)

SLR & Geodesy References

Pearlman, M. R., Degnan, J. J., & Bosworth, J. M. (2002). *The International Laser Ranging Service*. Advances in Space Research, 30(2), 135-143. DOI: 10.1016/S0273-1177(02)00277-6

Marini, J. W., & Murray, C. W. (1973). *Correction of Laser Range Tracking Data for Atmospheric Refraction at Elevations Above 10 Degrees*. NASA Technical Report X-591-73-351.

Mendes, V. B., & Pavlis, E. C. (2004). *High-accuracy zenith delay prediction at optical wavelengths*. Geophysical Research Letters, 31(14). DOI: 10.1029/2004GL020308

International Laser Ranging Service (ILRS). (2025). *SLR monthly normal point data*. Greenbelt, MD, USA: NASA Crustal Dynamics Data Information System (CDDIS). DOI: 10.5067/SLR/slr_data_monthly_npt_001

International Laser Ranging Service (ILRS). (2025). *SLR Combination Center (CC) Orbit Product*. Greenbelt, MD, USA: NASA Crustal Dynamics Data Information System (CDDIS). DOI: 10.5067/SLR/SLR_ILRSORB_001

International Laser Ranging Service (ILRS). (2025). *Satellite Laser Ranging Frame 2020 (SLRF2020) Product*. Greenbelt, MD, USA: NASA Crustal Dynamics Data Information System (CDDIS). DOI: 10.5067/SLR/slrf2020_001